

Chapter 1, Problem 33CP

Problem

A lightning bolt carried a current of 2 kA and lasted for 3 ms. How many coulombs of charge were contained in the lightning bolt?

Step-by-step solution

Step 1 of 1

Charge Contained in the Lightning Bolt

$$Q = I t$$

$$\text{Given } I = 2 \times 10^3 \text{ A} , t = 3 \times 10^{-3} \text{ s}$$

$$\therefore Q = 2 \times 10^3 \times 3 \times 10^{-3}$$

$$= 6 \text{ C}$$